CS 5306 / INFO 5306
Fall 2026

Project 2

The goal of this project is for you to explore what actionable "wisdom" may be embedded in the digital records of people's activities that we might find in various online sources. The goal is not to discover knowledge about the system whose data you're using, but rather to discover some hidden signal in that data that is relevant in some other setting. As examples you can review the papers on this topic that we've read in class, as well as the various additional papers summarized in the lecture slides.

Your goal is to think up intriguing possible connections between people's online behavior in one context and a question relevant in some other setting. I want you to ask a clear question, then answer it empirically from data. It's ok if you find that you either were wrong about what you expected or the results were simply equivocal or otherwise unable to answer the question. There are numerous sources of data you might consider, such as Google Trends, Twitter, Reddit, etc.

It is crucial that you think carefully about data. You must have some source data that reflects people's online behavior, and some signal that you believe you can predict from it. In some cases the system of interest maintains their own datasets for use by researchers and students or provides an API for access to such data. For example:

- Meta/Facebook provides various datasets and APIs for researchers at https://transparency.meta.com/researchtools, but you have to apply and be approved.
- Reddit: There are archives at https://github.com/ArthurHeitmann/arctic_shift and https://academictorrents.com/details/ba051999301b109eab37d16f027b3f49ade2de13.
- StackExchange provides data dumps via the information at https://meta.stackexchange.com/questions/224873/all-stack-exchange-data-dump-releases/224922#224922.
- X/Twitter:
    - There are online archives, such as TweetsKB, https://data.gesis.org/tweetskb/.
    - You can also try scraping the data yourself, and there are tools for this such as Altimis/Sctweet, https://github.com/Altimis/Scweet.
- There are also many datasets available at Kaggle, https://www.kaggle.com/datasets, and Hugging Face, https://huggingface.co/datasets.

What should you predict? There are many possibilities! But you need data for it. One possible direction is to use datasets linked to here in the context of "biothreats", https://www.hiddensignalschallenge.com/additional-resources/#data-resources.

There are three "deliverables" for this assignment:

Project proposal: Friday, October 9, 11:59pm
This should be ½-1 page, and should have the following sections:

1. The names and netIDs of the students on your team.
2. The system or data resource in which you are looking for the "wisdom of crowds". Please be specific – show us that there is data available.
3. The question you are asking – what you are hoping to show from the data.

Your proposal will not be graded other than whether or not it is submitted. Its purpose is to give you feedback on your proposed project if we believe it necessary. If we have questions or concerns that warrant it we may ask you for a revision to your proposal. If we do it does not reflect a "grade", but is instead simply our effort to have you gain maximum value from the project.

- **One-slide summary: Sunday, November 8, 10:00pm**
  This is the same as for Project 1.

- **Report: Monday, November 9, 10:00am**
  Your final submission will be a report of ~5 pages. This length is a guideline. Go longer if you need to include, for example, images or graphs. Your report should have the following sections in the following order:
  1. Title
  2. Names and NetIDs of teammates
  3. The online data/resource(s) used
  4. The question you originally had, and if it morphed why it changed and what you ended up asking
  5. Describe the conclusion you reached and the experimental basis you have for that conclusion.

  As with Project 1, you do not need to be correct in your prediction. You will be graded on the thoughtfulness and clarity of your question and investigation. Demonstrated effort also plays a significant factor in your grade.

Your investigation might be totally original, or it might replicate work that has already been done – under the presumption that some time has passed since the earlier study, so the answer might now be different. Your goal is to ask a question and use data that you gather to answer the question as best you can. You are not being graded on whether your conjectured answered to the question was right. It might very well be that what you found contradicted what you expected or was equivocal. What's important is being clear about the question you're asking, how you're answering it, and what answer you found.

A few words of warning:

- **Make sure you spend time making sure you have the data that you need for your project before submitting your proposal. This is the most important element of your proposal.**
- This isn't a project about programming. If part of your effort involved writing code, that's fine. You can use whatever language you like. You don't need to submit it. The project is about the question you are asking and the answer you get, and I assume you have the capacity to do what you need computationally to accomplish this.